413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 8

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1 Hypothesis Testing: A Recap

- Does a new treatment improve patient recovery?
- Does tobacco exposure change the level of a biological marker?
- Do two populations have different average outcomes or can the difference be explained by random fluctuations arising from the finite number of samples ?
- Is a coin fair, or are certain outcomes more likely?

In each case, we collect observations to evaluate a hypothesis about the underlying population or process. However, observed data are subject to random fluctuations. Even if a treatment has no effect, two independently sampled groups will generally not have exactly the same sample mean. Similarly, a fair coin will not necessarily produce equal numbers of heads and tails in an experiment with finite number of trials.

The central question of hypothesis testing is therefore:

Are the observed differences consistent with random fluctuations under a proposed hypothesis, or are they sufficiently unusual (large or small) to reject that hypothesis?

In much simpler words,

What is the probability that the observed difference was due to a fluke, or was there a systematic change?

We formulate a null hypothesis H_0, specifying a reference model for the observations, and an alternative hypothesis H_1, describing the departure from H_0 that we wish to investigate.

We then follow three steps:

1. **Formulate the hypotheses:** Specify H_0 and H_1, and assume H_0 is true.
2. **Construct a test statistic:** Choose a quantity that measures the discrepancy between the observations and the predictions of H_0. Determine the probability distribution under H_0.
3. **Evaluate the observations:** Determine how unusual the observed statistic is under H_0 and compare its p-value with a pre-specified significance level α.

1.1 Example: Is a Coin Biased?

Suppose we flip a coin $n = 20$ times and observe 14 heads. We want to determine whether the coin is biased toward heads.

Step 1: Formulate the hypotheses. Let p denote the probability of heads. We test

$$H_0 : p = 0.5, \qquad H_1 : p > 0.5.$$

Step 2: Construct a test statistic. Let H denote the number of heads in 20 flips. Under H_0, assuming independent flips,

$$\boxed{H \sim \text{Binomial}(20, 0.5).}$$

We choose the test statistic

$$\boxed{T = H.}$$

Since $H_1 : p > 0.5$, unusually large values of T provide evidence against H_0.

Step 3: Compute the p-value. We observed $t_{\text{obs}} = 14$. The probability of observing at least 14 heads under H_0 is

$$\begin{aligned} p\text{-value} &= P(T \geq 14 \mid H_0) \\ &= \sum_{k=14}^{20} \binom{20}{k} (0.5)^k (0.5)^{20-k} \\ &= \sum_{k=14}^{20} \binom{20}{k} (0.5)^{20} \\ &\approx 0.0577. \end{aligned}$$

Since $0.0577 > 0.05$, we fail to reject H_0 at significance level $\alpha = 0.05$.

Key point. A test statistic is a function of the observed data whose probability distribution under H_0 allows us to determine how unusual the observation is. It need not follow a Gaussian distribution.

Different scientific questions and experimental designs require different test statistics. We now introduce some commonly used statistics and discuss when each is appropriate.

1.2 The Z-Statistic

Suppose we collect n independent and identically distributed observations $X_1, \ldots, X_n$ with

$$\mathbb{E}[X_i] = \mu, \qquad \text{Var}(X_i) = \sigma^2.$$

Suppose the null hypothesis specifies a population mean μ_0, i.e.,

$$H_0 : \mu = \mu_0.$$

We use the sample mean $\bar{X}_n$ to assess whether the observed data are consistent with this hypothesis. To assess the discrepancy between the observed sample mean and the prediction under H_0, recall that

$$\mathbb{E}[\bar{X}_n] = \mu, \qquad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

When the population standard deviation σ is known, we construct the standardized test statistic

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}.$$

The numerator measures the deviation from H_0, while the denominator quantifies the expected sampling fluctuations. Under H_0,

$$Z \sim \mathcal{N}(0, 1)$$

exactly for Gaussian observations, and approximately for sufficiently large samples by the Central Limit Theorem. Thus, the Z-statistic measures how unusual the observed sample mean is relative to its distribution under H_0.

1.3 One-Sided and Two-Sided Alternatives

The alternative hypothesis determines which values of the test statistic are considered extreme.

Alternative hypothesis	p-value for observed z_{obs}
$H_1 : \mu > \mu_0$	$P(Z \geq z_{\text{obs}})$
$H_1 : \mu < \mu_0$	$P(Z \leq z_{\text{obs}})$
$H_1 : \mu \neq \mu_0$	$P(\lvert Z\rvert \geq \lvert z_{\text{obs}}\rvert)$

For a two-sided test, using the symmetry of the standard normal distribution,

$$\boxed{p = 2\left[1 - \Phi(|z_{\text{obs}}|)\right].}$$

At a pre-specified significance level α,

$$\boxed{\begin{cases} p \le \alpha & \Longrightarrow \text{Reject } H_0, \\ p > \alpha & \Longrightarrow \text{Fail to reject } H_0. \end{cases}}$$

1.4 Example: Does Tobacco Consumption good or bad ?

Suppose a biomarker which has proven to be accurate indicator for lung capacity has a known population mean $\mu_0 = 50$ and standard deviation $\sigma = 10$ in non-smoking population. For instance, smaller values of the biomarker indicate decreased lung capacity.

We then collect independent measurements from $n = 100$ smoking individuals and observe a sample mean

$$\bar{X}_n = 48.2.$$

We would like to determine whether tobacco consumption is associated with a change in the population mean of this biomarker.

Step 1: Formulate the hypotheses. Consider two different scientific questions.

Question A: Is the mean biomarker level lower? We use a one-sided alternative:

$$H_0 : \mu = 50, \qquad H_1 : \mu < 50.$$

Question B: Is the mean biomarker level different? We use a two-sided alternative:

$$H_0 : \mu = 50, \qquad H_1 : \mu \neq 50.$$

The first test considers decrease in the biomarker reading, whereas the second considers changes in either direction. For instance, the second considers a scenarios tobacco is good for you.

Step 2: Construct the test statistic.
For both scenarios, we use the same Z-statistic:

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}.$$

Under H_0, assuming Gaussian measurements or a sufficiently accurate CLT approximation,

$$Z \approx \mathcal{N}(0, 1).$$

The observed value is

$$\boxed{z_{\text{obs}} = \frac{48.2 - 50}{10/\sqrt{100}} = -1.8.}$$

Thus, the observed sample mean is 1.8 standard errors below the value predicted by H_0.

Step 3: Compute the p-values.

Question A: One-sided test. Since $H_1 : \mu < 50$, we consider the left tail:

$$\begin{aligned} p_{\text{one}} &= P(Z \le -1.8 \mid H_0) \\ &= \Phi(-1.8) \\ &\approx 0.0359. \end{aligned}$$

At significance level $\alpha = 0.05$,

$$p_{\text{one}} = 0.0359 < 0.05.$$

Therefore,

$$\text{Reject } H_0.$$

Question B: Two-sided test. Since $H_1 : \mu \neq 50$, deviations in either direction are relevant:

$$\begin{aligned} p_{\text{two}} &= P(|Z| \ge 1.8 \mid H_0) \\ &= 2[1 - \Phi(1.8)] \\ &\approx 0.0718. \end{aligned}$$

At significance level $\alpha = 0.05$,

$$p_{\text{two}} = 0.0718 > 0.05.$$

Therefore,

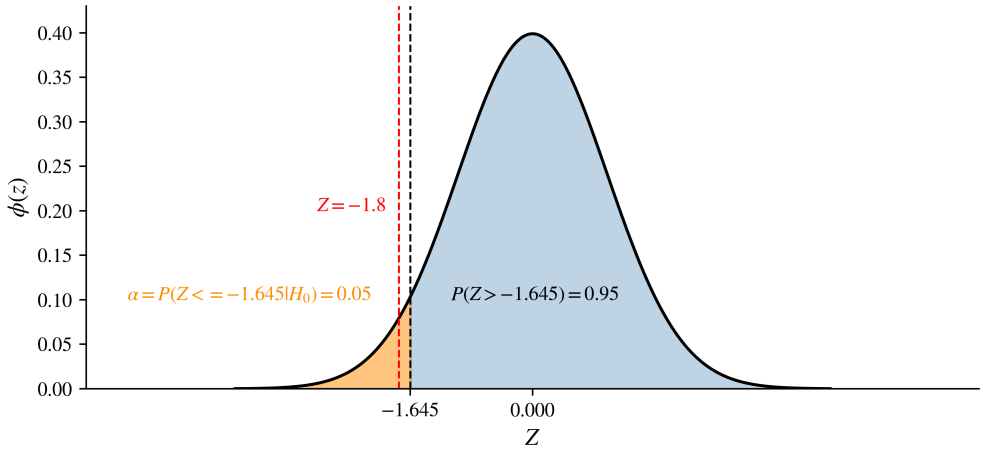
$$\text{Fail to reject } H_0.$$

What changed? Both tests used the same observations and the same test statistic:

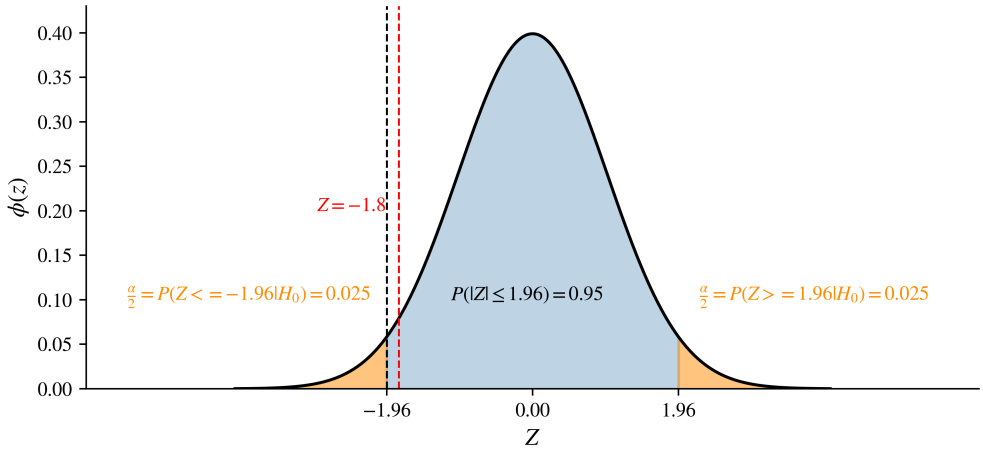
$$z_{\text{obs}} = -1.8.$$

However, they used different rejection regions:

Test	Critical value(s) at $\alpha = 0.05$	Decision
One-sided	$Z \leq -1.645$	Reject H_0
Two-sided	$\lvert Z \rvert \geq 1.96$	Fail to reject H_0



One-sided test: $z_{\text{obs}} = -1.8 < -1.645$, so the observed statistic lies in the rejection region. Therefore, we reject H_0 at $\alpha = 0.05$.



Two-sided test: $|z_{\text{obs}}| = 1.8 < 1.96$, so the observed statistic does not lie in the rejection region. Therefore, we fail to reject H_0 at $\alpha = 0.05$.

Figure 1: Comparison of one-sided and two-sided hypothesis tests for the same observed statistic $z_{\text{obs}} = -1.8$.

Important lesson: One-sided vs. two-sided tests and p-hacking.
The alternative hypothesis determines which observations count as extreme under H_0. A one-sided test places the rejection region in one pre-specified direction, whereas a two-sided test considers deviations in either direction.

In our tobacco-exposure example, the same observations produced

$$p_{\text{one}} = 0.0359, \qquad p_{\text{two}} = 0.0718.$$

At a pre-defined significance level $\alpha = 0.05$, the one-sided test rejects H_0, whereas the two-sided test fails to reject H_0. Neither test is inherently incorrect: they address different scientific questions.

What is p-hacking? p-hacking refers to selecting or modifying statistical analyses after examining the data, particularly to obtain a statistically significant result, without adequately disclosing these choices. For instance, suppose a researcher initially planned to test

$$H_1 : \mu < 50,$$

but obtained $p = 0.0359 < 0.05$ and hence had to reject the null hypothesis H_0. So, the researcher changes the alternative to

$$H_1 : \mu \neq 50,$$

obtaining $p = 0.0718 > 0.05$, thereby, failing to reject the null hypothesis, and reports only this two-sided result.

This is an example of p-hacking: the hypothesis was chosen after inspecting the observations to obtain a favorable statistical conclusion. Conversely, a researcher could selectively choose a one-sided analysis, or emphasize its significant result, to exaggerate the claim. The concern is not that a two-sided test is inappropriate, but that choices are made or reported selectively to favor a desired conclusion.

Good scientific practice.

- Specify the primary hypotheses and statistical analysis before examining the data.
- Choose one-sided or two-sided tests based on the scientific question, not the observed p-value.
- Report changes to the analysis transparently and distinguish exploratory from pre-specified tests.
- Do not interpret failure to reject H_0 as proof that H_0 is true.

> Statistical significance depends not only on the observations, but also on the hypotheses and analysis chosen to evaluate them. These choices must be scientifically justified and transparently reported.

1.5 Student's t-Statistic: An Example

Suppose a manufacturing process is designed to produce components with a mean weight of 50 grams. To check whether the process has changed, we randomly sample $n = 16$ components and observe

$$\bar{X}_n = 52 \text{ g}, \qquad S_n = 4 \text{ g}.$$

Unlike in the previous example, the population standard deviation σ is unknown. We want to determine whether the mean weight has changed.

Step 1: Formulate the hypotheses. Let μ denote the population mean weight. We test

$$\boxed{H_0 : \mu = 50, \qquad H_1 : \mu \neq 50.}$$

Step 2: Construct a test statistic and determine its distribution under H_0. Recall that the sample mean has standard error

$$\text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

However, σ is unknown. We estimate it using the sample standard deviation

$$S_n = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}$$

which in itself is a random variable. This gives the estimated standard error

$$\widehat{\mathrm{SE}}(\bar{X}_n) = \frac{S_n}{\sqrt{n}}.$$

We therefore construct Student's t-statistic:

$$\boxed{T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}}.}$$

Unlike the Z-statistic, the denominator now contains the random variable S_n.

Note: S_n is a random variable because it depends on the observations. Drawing another random sample of size n will generally produce a different value of S_n.

If the observations are independent and normally distributed, then under H_0,

$$\boxed{T \sim t_{n-1}.}$$

Here, $n-1$ denotes the degrees of freedom. For our sample,

$$df = 16 - 1 = 15.$$

The observed statistic is

$$\begin{aligned} t_{\text{obs}} &= \frac{52 - 50}{4/\sqrt{16}} \\ &= 2 \end{aligned}$$

Thus, the observed sample mean is two estimated standard errors above the hypothesized mean.

Step 3: Determine how unusual the observation is under H_0.
Since $H_1 : \mu \neq 50$, we use a two-sided test:

$$p = P(|T_{15}| \geq 2) \approx 0.064.$$

Using the Student's t-table, for $df = 15$, the two-sided critical values are

Two-tail probability	0.10	0.05
Critical value	1.753	2.131

Since

$$1.753 < 2 < 2.131,$$

we obtain

$$0.05 < p < 0.10.$$

At significance level $\alpha = 0.05$,

$$p > 0.05.$$

Therefore,

$$\text{Fail to reject } H_0.$$

The observed difference is not sufficiently unusual under H_0 to reject it at the 5% significance level.

Why use a t-statistic instead of a Z-statistic? In the Z-test, the population standard deviation σ is known:

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}.$$

Here, σ is unknown and must be estimated:

$$T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}}.$$

Since S_n fluctuates between samples, the resulting statistic follows a Student's t-distribution rather than a standard normal distribution (exactly for independent Gaussian observations).

Student's t-distribution has heavier tails, reflecting the additional uncertainty from estimating the population standard deviation.

1.6 Welch's t-Statistic: Comparing Two Populations

Suppose we want to determine whether a new teaching method changes students' mean exam scores compared with a traditional teaching method. We collect two independent samples:

Group	Sample size	Sample mean	Sample standard deviation
Traditional (C)	$n_C = 25$	$\bar{X}_C = 70$	$S_C = 8$
New method (T)	$n_T = 25$	$\bar{X}_T = 76$	$S_T = 12$

Although both groups have the same sample size, their variability differs. We do not assume equal population variances.

Step 1: Formulate the hypotheses.
Let μ_C and μ_T denote the population mean scores. We test

$$\boxed{H_0 : \mu_T = \mu_C, \qquad H_1 : \mu_T \neq \mu_C.}$$

Step 2: Construct a test statistic and determine its distribution under H_0.
The observed difference between the sample means is $\bar{x}_T - \bar{x}_C = 6$. To evaluate this difference, we need its standard error.

For two independent samples,

$$\mathrm{Var}(\bar{X}_T - \bar{X}_C) = \frac{\sigma_T^2}{n_T} + \frac{\sigma_C^2}{n_C}.$$

This identity follows from independence and does not require Gaussian distributions or the CLT.

Since the population variances are unknown, we estimate them using the sample variances S_T^2 and S_C^2. This gives **Welch's t-statistic**:

$$T_W = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{S_T^2/n_T + S_C^2/n_C}}.$$

The denominator is the estimated standard error of the difference between the sample means. For equal sample sizes $n_T = n_C = n$, this simplifies to

$$T_W = \frac{\sqrt{n}\,(\bar{X}_T - \bar{X}_C)}{\sqrt{S_T^2 + S_C^2}}.$$

For our observations,

$$\begin{aligned} t_{\text{obs}} &= \frac{76 - 70}{\sqrt{12^2/25 + 8^2/25}} \\ &= \frac{6}{\sqrt{8.32}} \\ &\approx 2.08. \end{aligned}$$

To determine the distribution under H_0, assume that the observations in each group are independent and Gaussian (*this is an additional assumption about the data we are making*). Welch's test then uses an approximate Student's t-distribution. Its degrees of freedom depend on the sample sizes and variances; statistical software gives approximately 42 for this example.

Step 3: Determine how unusual the observation is under H_0.
Since $H_1 : \mu_T \neq \mu_C$, we use a two-sided test:

$$p \approx P(|T_{42}| \geq 2.08) \approx 0.044,$$

where T_{42} denotes a Student's t random variable with 42 degrees of freedom.

Using the nearby $df = 40$ row of the Student's t-table,

Two-tail probability	0.05	0.02
Critical value	2.021	2.423

Since

$$2.021 < 2.08 < 2.423,$$

we obtain

$$0.02 < p < 0.05.$$

At significance level $\alpha = 0.05$,

$$\text{Reject } H_0.$$

The observed difference is sufficiently unusual under H_0 to reject equality of the population means at the 5% significance level.

When to use Welch's t-test.
Welch's test compares the means of two independent populations without assuming equal population variances. The estimated standard error accounts for sampling fluctuations in both groups:

$$\widehat{\text{SE}} = \sqrt{\frac{S_T^2}{n_T} + \frac{S_C^2}{n_C}}.$$

The Student's t reference distribution is approximate, even for Gaussian populations with unequal variances. For sufficiently large non-Gaussian samples with finite variances, the CLT provides an asymptotic standard normal approximation. Neither approximation is guaranteed to be accurate for small, strongly non-Gaussian samples.

2 Parameter Estimation and Distribution Fitting

2.1 From Probability to Statistics

Until now, we have mostly worked in the following setting: the probability distribution of a random variable was known, and its parameters were also known. For example,

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

with specified values of μ and σ^2, or

$$X \sim \text{Poisson}(\lambda)$$

with a known value of λ. In that setting, we used the distribution to answer questions such as

$$P(X \leq x), \qquad \mathbb{E}[X], \qquad \text{Var}(X).$$

In statistics, the situation is usually reversed. We observe data

$$X_1, \ldots, X_n,$$

but the parameters of the underlying distribution are unknown. For example, we may assume

$$X_i \sim \text{Poisson}(\lambda),$$

but λ is unknown, or

$$X_i \sim \mathcal{N}(\mu, \sigma^2),$$

but μ and σ^2 are unknown.

The goal is now to use the observed data to estimate these unknown parameters:

$$\boxed{X_1, \ldots, X_n \quad \longrightarrow \quad \hat{\theta}.}$$

This is the problem of **parameter estimation**. Once $\hat{\theta}$ is estimated, the fitted model

$$p_{\hat{\theta}}(x)$$

provides a probabilistic description of the data.

Probability: known parameters $\longrightarrow$ generate data

Statistics: observed data $\longrightarrow$ unknown parameters

2.2 Classical Examples

For many familiar distributions, θ contains only a few parameters:

Distribution	Parameters
Poisson	$\theta = \lambda$
Gaussian	$\theta = (\mu, \sigma^2)$
Exponential	$\theta = \lambda$

For example, if

$$X \sim \text{Poisson}(\lambda),$$

then

$$\mathbb{E}[X] = \lambda.$$

Hence, from observations $X_1, \ldots, X_n$, a natural estimate is

$$\hat{\lambda} = \bar{X}_n.$$

This is an example of the **method of moments**: theoretical moments of the distribution are matched with empirical moments computed from the data.

2.3 Modern Machine Learning

The same idea extends to much more complicated probability models. Instead of a distribution with one or two parameters, we may use a neural network with parameters

$$\theta = (\theta_1, \ldots, \theta_P),$$

where P can be extremely large. For instance, the latest GPT-6 Astra is estimated to have $\approx 1 - 5$ trillion parameters. Training estimates these parameters from very large datasets, and large computational resources. The underlying statistical idea is the same: choose parameters so that the model provides a good description of the observed data.

After training, the parameters $\hat{\theta}$ are fixed. During inference, the model uses the learned probability distribution

$$p_{\hat{\theta}}(x_{\text{next}} \mid x_{\text{context}})$$

to generate new outputs, typically by sampling from this distribution.

> Parameter estimation is the problem of learning the unknown parameters of a probability model from data. The number of parameters may range from one, as in a Poisson model, to trillions or more in large language models.

There are several ways to estimate unknown parameters from data. Two common approaches are:

- **Method of Moments (MoM):** match theoretical moments of the model with sample moments computed from the data.
- **Maximum Likelihood Estimation (MLE):** choose the parameter values that make the observed data most likely under the model.

In this lecture, we begin with the method of moments because it is simple, intuitive, and often leads to closed-form estimators.

2.4 Method of Moments

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} p_\theta(x),$$

where the form of the distribution is known, but the parameter θ is unknown. The goal is to estimate θ from the data.

The **method of moments** uses the moments of the distribution,

$$\mu_k = \mathbb{E}[X^k],$$

and matches them with the corresponding sample moments,

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

If the parameter can be written in terms of the population moments,

$$\theta = g(\mu_1, \mu_2, \ldots),$$

we estimate it by replacing the population moments with their sample counterparts:

$$\boxed{\hat{\theta}_{\mathrm{MM}} = g(\hat{\mu}_1, \hat{\mu}_2, \ldots).}$$

2.5 Example 1: Poisson Distribution

Suppose

$$X_i \sim \mathrm{Poisson}(\lambda),$$

where λ is unknown, but we know the structure of the distribution (Poisson). For a Poisson random variable,

$$\mathbb{E}[X] = \lambda.$$

Therefore,

$$\lambda = \mu_1.$$

Replacing the population moment by the sample moment gives

$$\boxed{\hat{\lambda}_{\mathrm{MM}} = \hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n.}$$

Thus, the method-of-moments estimate of the Poisson rate is simply the sample mean.

2.6 Example 2: Gaussian Distribution

Suppose

$$X_i \sim \mathcal{N}(\mu, \sigma^2),$$

with both μ and σ^2 unknown. The first two moments satisfy

$$\mu_1 = \mathbb{E}[X] = \mu,$$

and

$$\mu_2 = \mathbb{E}[X^2] = \sigma^2 + \mu^2.$$

Hence,

$$\mu = \mu_1, \qquad \sigma^2 = \mu_2 - \mu_1^2.$$

Replacing the population moments by sample moments gives

$$\hat{\mu}_{\mathrm{MM}} = \bar{X}_n,$$

and

$$\hat{\sigma}^2_{\mathrm{MM}} = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}_n^2.$$

The method of moments estimates unknown parameters by matching **population moments** predicted by the model with **sample moments** computed from the data:

$$\boxed{\text{population moments} \longrightarrow \text{sample moments.}}$$

For large samples, these sample moments approach the corresponding population moments, motivating the resulting parameter estimates.